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## 1 Executive Summary

Founded in 1927, Marriott International is a global leader in hospitality, operating over 8,800 properties across 30+ brands spanning luxury, premium, and lifestyle segments. As the travel industry rebounds post-pandemic, Marriott aims to strengthen its competitive edge by aligning its services with evolving guest expectations, optimizing brand loyalty, and capitalizing on emerging market opportunities.

This study is structured to address Marriott's strategic priorities through six research aims which are understanding customer preferences for personalized services and technology, in order to deliver exceptional guest experience based on different demographics segments, identifying emerging trends and destinations hotspots, assess guest satisfaction across different brands and regions, explore the impact of sustainability initiatives on customer loyalty, evaluate the effectiveness of customer loyalty programs and understand how economic factors influence travel behaviors could help the company adapt to changing market conditions.

Research Aims (RAs) will be turned into Research Questions (RQs) and Research Objectives (ROs). These will guide us in collecting and analyzing data on various topics like customer preferences, brand loyalty, market trends, sustainability, and economic factors. This ensures the research aligns with Marriott's goals. This proposal explains the method we will use to collect data, including the types of information we need and the methods we'll use to analyze it. We will also describe how we will select people for the survey and what data Marriott will provide to make the research more accurate. Additionally, we will outline a timeline and budget to keep the project on track and within budget. Finally, we will include a Gantt chart and a cost breakdown to show the project's stages and help Marriott understand the costs and deadlines. Overall, the research will provide Marriott with useful recommendations to meet its goals of delivering great guest experiences and seizing new opportunities in the hospitality market. By understanding customer preferences and market trends, Marriott will be in a better position to stay competitive, boost loyalty, and respond to changes in the market.

## 2 Background

As one of the leaders in global hospitality, Marriott International was founded in 1927. It has over 30 hotel brands and 10,000 global destinations. During pandemic, Marriott Bonvoy program is used to improve guest loyalty through initiatives after company recognizes the need to meet evolving customer expectations. The company is focused on improving its services to meet the changing needs of its guests and adapt to shifting market conditions.

## 3 Problem Definition

The client brief mentioned Marriott's business objectives in increasing its hotel properties and improving brand diversity. To achieve these goals, Marriott aims to gain deeper insights into the factors that influence guests' preferences . Furthermore, the company also aims to use technology to meet evolving market demands. This study will focus on these issues over the next 6 months and suggest ways for Marriott to improve its services and stay competitive in the hospitality market

## 4 Research Aims (RAs)

Based on the client brief, we have identified the following research aims to address Marriott International's concerns:

- Research Aim 1: Understanding customer preferences for personalized services and technology, in order to deliver exceptional guest experience based on different demographics segments.
- Research Aim 2: Identifying emerging trends and destinations hotspots.
- Research Aim 3: Assess guest satisfaction across different brands and regions.
- Research Aim 4: Explore the impact of sustainability initiatives on customer loyalty.
- Research Aim 5: Evaluate the effectiveness of customer loyalty programs.

- Research Aim 6: Understand how economic factors influence travel behaviors could help the company adapt to changing market conditions.

## 5 Research Questions (RQs)

For each research aim, we have come up with some research questions (RQs), listed below:

1) For RA1:

RQ 1: Do customer preferences for personalized services differ between Marriott customers and non-Marriott customers?

2) For RA2:

RQ2: What factors distinguish between destinations perceived as affordable and those perceived as less affordable by travelers?

3) For RA3:

RQ3: How do different hotel brands and regions affect guest satisfaction levels?

4) For RA4:

RQ4: Do different age groups have different levels of awareness about Marriott's sustainability initiatives?

5) For RA5:

RQ5: What are the key factors that influence customers to join the Bonvoy rewards program?

6) For RA6:

RQ6: How do economic factors influence the annual number of international bookings at Marriott hotels?

## 6 Research Objectives (ROs)

For each research question, we have created research objectives to answer them and meet our client's goals.

1) For RQ1:

RO1: To examine how customer preferences for personalized services differ between Marriott customers and non-Marriott customers.

2) For RQ2:

RO2: Determine the key factors that influence travelers' perceptions of a destination's affordability.

3) For RQ3:

RO3: To determine how guest satisfaction varies significantly across different hotel brands and regions

4) For RQ4:

RO4: To compare sustainability awareness among different age groups.

5) For RQ5:

RO5: Identify the main reasons why people join the Bonvoy rewards program.

6) For RQ6:

RO6: To examine the impact of factors on the annual number of international bookings at Marriott hotels.

# 7 Methodology

## 7.1 Research Design

The market objectives incorporate exploratory and descriptive designs in alignment with the aims of the research. Exploratory research helps us to gain knowledge to understand problems and issues brought about by the respondents. The descriptive design is used to gain an understanding about customers' level of satisfaction towards personalized services in segments without aiming to change or affect their behavior.

We are using a mixed mode in this research design. For qualitative analysis, the research uses focus groups to assess the effectiveness of the Marriott Bonvoy rewards program. For quantitative analysis, we will develop hypotheses for each research objective and use data to find relationships between the variables.

We would be collecting the primary data from the questionnaire and focus group while secondary data will be sourced from Marriott's marketing database to enhance the understanding of customer behavior. Incentives will be provided to participants to ensure a higher response rate and encourage cooperation.

Quantitative:

RO1 – Independent sample t-test

RO2 - 3-group discriminant analysis

RO3 – 2-way ANOVA

RO4 – 1-way ANOVA

RO6 – Multiple linear regression

Qualitative:

RO5 – Focus group



## 8 Data Collection

### 8.1 Factors to consider for choosing respondents for the study

This study will focus on adults aged 18 and older, as they don't need parental consent to participate. Adults are likely to have a clearer understanding of topics like customer loyalty programs and personalized services, making them better suited to provide meaningful feedback. Since the study uses online questionnaires and focus groups, it is important that participants are familiar with basic technology. Respondents who regularly use email are likely to have the necessary computer skills to complete the study, ensuring that the data we collect is accurate and reliable. Additionally, having access to a computer or smartphone will make it easier for participants to take part in the study and share their thoughts effectively.

### 8.2 Questionnaire

To sort customers by various demographics in minimal time, online questionnaires should be used to avoid cumbersome paperwork. We should have an overall section with general questions to be answered by every customer, including gender, age, accommodation price range, and price range for travel. This shall be followed by taking customers to various sections in accordance with their status (Marriott/non-Marriott) in response to their answers. As a final point, in case they opt to join the focus group, an interest form shall be completed by them.

We will conduct a pilot test with 30 respondents to identify and fix any issues in the questionnaire. The final questionnaire will be clear and easy for respondents, with no confusion about the content, wording, order, format, difficulty, or instructions.

#### 8.2.1 Selecting sample size for Questionnaire

We have approximately 5,000 Marriott customers. We have used a confidence level of 95% (5% significance) and a conservative proportion of 0.5. We apply the formula below to calculate the sample required:

$$n \geq \frac{Z_{\alpha/2}^2(\pi(1 - \pi))}{e^2}$$

$$n \geq \frac{1.96^2(0.5(1 - 0.5))}{0.05^2} \approx 385$$

The average response rate is 44.1%<sup>1</sup>.

We may achieve a minimum sample size of 873.

Since the exact population size of non-Marriott customers is unknown, we assume that the minimum required sample size remains 873, considering the population is sufficiently large.

### 8.2.2 Questionnaire involving Marriott customers

We divide Marriott customers into groups based on booking habits or membership tiers (e.g., Silver, Gold, Platinum) using Marriott's data. For each group, we randomly select people to survey. This method will make results more accurate and ensure all customer types are included fairly.

### 8.2.3 Questionnaire involving non-Marriott customers

We use a quick and low-cost method to select respondents. We will share the survey on social media and travel websites and offer a free hotel stay as a prize for sharing the survey publicly. This helps collect many responses fast, but results may be less reliable than random sampling.

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<sup>1</sup> Quoted from Response rates of online surveys in published research: A meta-analysis

## 8.3 Focus groups

We will host online focus groups with 50 Marriott customers (selected from survey who sign-ups) to understand why they join the Bonvoy rewards program. Groups of 10 participants each will meet for 90 minutes on Zoom, split over 2 weeks. To get honest feedback, we mix members from different loyalty tiers (Silver, Gold, Platinum) and use simple visuals (e.g., reward charts) to spark deeper talks. Participants join from home for comfort, and each gets a \$25 Amazon voucher after the session. If too many sign up, we'll pick frequent travelers first. This method saves time, avoids travel hassles, and highlights key loyalty drivers (e.g., instant rewards, status perks, or emotional ties to Marriott).

## 8.4 Flow chart

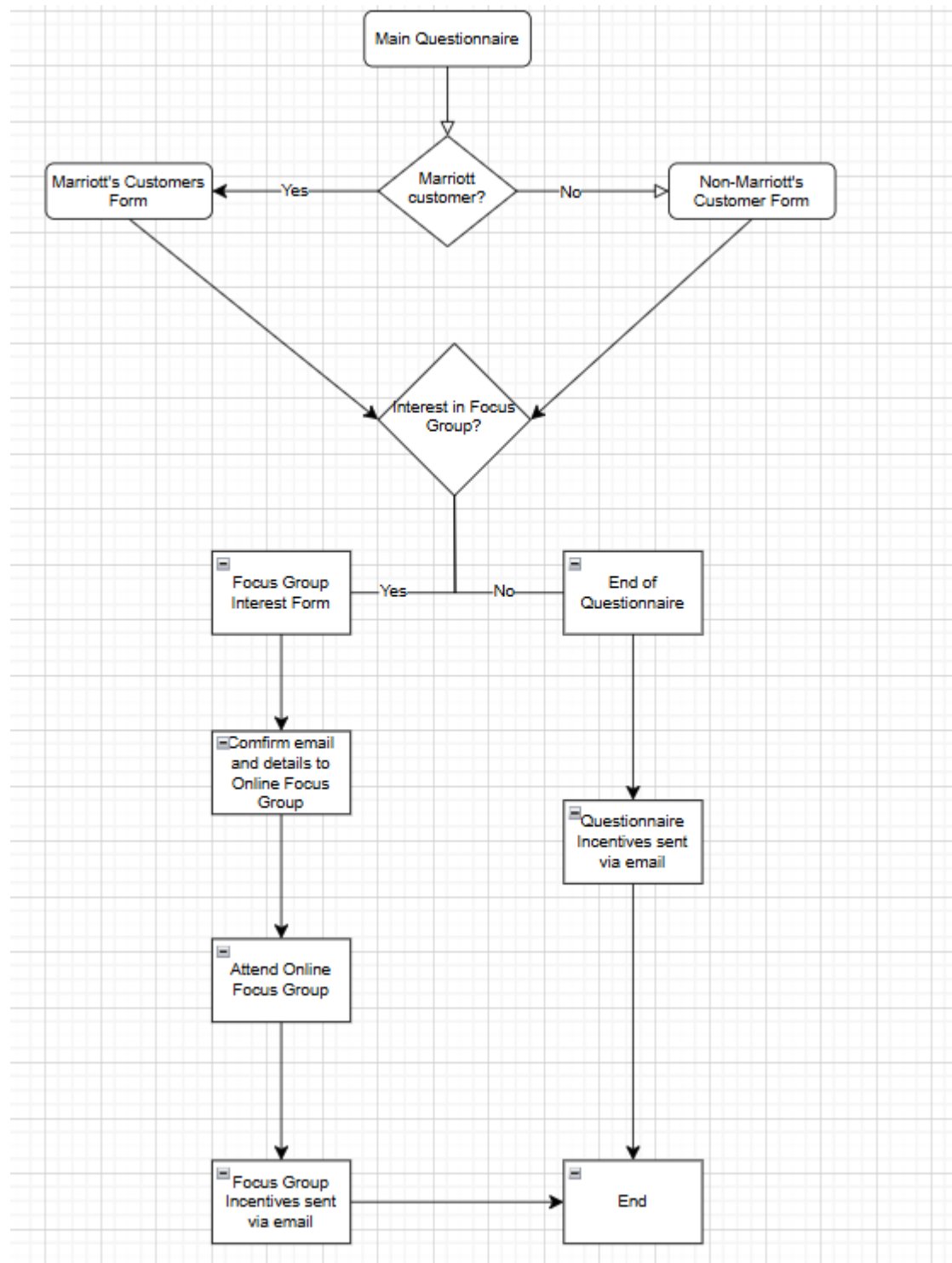


Figure 1: Flow chart to illustrate the start and end of the data collection process using questionnaire and Focus Group Discussion

## 8.5 Data Collected

| Variable                                                      | Data Types                     |
|---------------------------------------------------------------|--------------------------------|
| Gender                                                        | Categorical Nominal            |
| Age                                                           | Continuous/Categorical Ordinal |
| Region of Residence                                           | Categorical Nominal            |
| Customer Type                                                 | Categorical Nominal            |
| Preferences for personalized service                          | Continuous                     |
| Affordability                                                 | Categorical Ordinal            |
| Average Accommodation Cost                                    | Continuous                     |
| Flight Availability and Cost                                  | Continuous                     |
| Currency Exchange Rate                                        | Continuous                     |
| Guest satisfaction                                            | Continuous                     |
| Hotel Brand                                                   | Categorical Nominal            |
| Sustainability Awareness Score                                | Continuous                     |
| Annual number of international bookings at<br>Marriott hotels | Continuous                     |
| Travel Restrictions                                           | Categorical Ordinal            |

*Table 1: List of Data Collected*

## 9 Data Analysis

### 9.1 Quantitative

All tests will be conducted in SPSS using a 95% confidence interval and a 5% significance level

#### 9.1.1 Research Objective 1

An independent sample t test will be used for RO1. We will treat customer type as the independent variable and preferences for personalized services as the dependent variable to test if Marriott customers and non-Marriott customers have different preferences for personalized services.

Since sample sizes are large enough for both groups, the normality assumptions can be safely ignored required for this statistical test. Variables to be used for the test:

| Symbol       | Denotes                                                                                 |
|--------------|-----------------------------------------------------------------------------------------|
| $\sigma_1^2$ | Population variance of preference for personalized services of Marriott customer        |
| $\sigma_2^2$ | Population variance of preference for personalized services of non-Marriott customer    |
| $\mu_1$      | Population <u>mean</u> of preference for personalized services of Marriott customer     |
| $\mu_2$      | Population <u>mean</u> of preference for personalized services of non-Marriott customer |

*Table 2: Variable denotation for independent T-test*

Output from SPSS:

### Independent Samples Test

|                                      |                             | Levene's Test for Equality of Variances |      | t-test for Equality of Means |    |                 |                 |                       |                                           |       |
|--------------------------------------|-----------------------------|-----------------------------------------|------|------------------------------|----|-----------------|-----------------|-----------------------|-------------------------------------------|-------|
|                                      |                             | F                                       | Sig. | t                            | df | Sig. (2-tailed) | Mean Difference | Std. Error Difference | 95% Confidence Interval of the Difference |       |
| Preference for personalized services | Equal variances assumed     |                                         | *    |                              |    | #               |                 |                       | Lower                                     | Upper |
|                                      | Equal variances not assumed |                                         |      |                              |    | ##              |                 |                       |                                           |       |

Table 2: Independent samples T-test from SPSS

### Levene's Test

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

Reject  $H_0$  if value at \* < 0.05. This concludes that the population variances of preference for personalized services are significantly different for both Marriott customers and non-Marriott customers.

### Independent sample T-test

$$H_0: \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 - \mu_2 \neq 0$$

Reject  $H_0$  if the value in either # or ## (according to Levene's Test) is below 0.05. This indicates that population mean of preference for personalized services is significantly different for both Marriott customers and non-Marriott customers. If both the lower and upper bounds of the 95% confidence interval are positive, it suggests that Marriott customers have a stronger preference for personalized services compared to non-Marriott customers.

## 9.1.2 Research Objectives 2

To address this objective, we will use 3 group Discriminant Analysis. Let the predictor be the average accommodation cost, flight availability and cost, and currency exchange rate. The dependent variable is the destination's affordability which are low affordable, moderate affordable and high affordable.

Theoretical model:

$$D_1 = \alpha_0 + \alpha_1 AAC + \alpha_2 FAC + \alpha_3 CER$$

$$D_2 = \beta_0 + \beta_1 AAC + \beta_2 FAC + \beta_3 CER$$

The estimated model:

$$\widehat{D}_1 = \widehat{\alpha}_0 + \widehat{\alpha}_1 AAC + \widehat{\alpha}_2 FAC + \widehat{\alpha}_3 CER$$

$$\widehat{D}_2 = \widehat{\beta}_0 + \widehat{\beta}_1 AAC + \widehat{\beta}_2 FAC + \widehat{\beta}_3 CER$$

$D$ : Discriminant Score

$\alpha_k, \beta_k$ : Discriminant function coefficient for variable  $k$ ,  $k=1,2,3$

AAC: Average Accommodation Cost

FAC: Flight Availability and Cost

CER: Currency Exchange Rate

**Canonical Discriminant Function Coefficients**

|                              | Function 1 | Function 2 |
|------------------------------|------------|------------|
| Average Accommodation Cost   |            |            |
| Flight Availability and Cost |            |            |
| Currency Exchange Rate       |            |            |
| (Constant)                   |            |            |

*Table 3: Canonical Discriminant Function Coefficients from SPSS*

The coefficients in the discriminant function can be extracted from Table 3, whereas each significance of the predictors will be established by a series of ANOVAs with Wilks's



Lambda and P-values (tabulated in Table 4). From the SPSS result, 2 functions were extracted from the Discriminant Analysis procedure. Function in Wilks' Lambda table can be removed from further consideration if p-value >0.05, it does not provide additional discriminatory power.

**Test of Equality of Group Means**

|                                    | Wilks's<br>Lamda | F | df1 | df2 | Sig. |
|------------------------------------|------------------|---|-----|-----|------|
| Average<br>Accommodation<br>Cost   |                  |   |     |     |      |
| Flight<br>Availability and<br>Cost |                  |   |     |     |      |
| Currency<br>Exchange Rate          |                  |   |     |     |      |

*Table 4: Tests of Equality of Group Means from SPSS*

Predictors are statistically significant if p-value <0.05. In Table 4, a smaller Wilks' Lambda value or a larger F value indicates that the predictor has better discriminatory power in the model. Structure Matrix presents canonical correlations and provides the correlation between each predictor<sup>2</sup>.

To assess the model's fit, the Eigenvalues table shows how well the model discriminates between the three groups. In the Wilks' Lambda table, if the p-value is less than 0.05, we conclude that the discriminant function is effective at distinguishing between the groups. If the p-value is greater than 0.05, the predictor can be removed because it doesn't add discriminatory value.

Combined group plot shows more variation of group centroid across two functions. If the centroids are well-separated horizontally or vertically, it shows that function does the heavy lifting in discriminating between the three groups.

Using the 'Classification Results,' determine the hit ratio based on the validation sample. Since a hit ratio of 33.3% would be expected by chance alone for three groups, we will

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<sup>2</sup> Higher correlations - better predictor variables

calculate the improvement over this baseline. The result gives us insight into whether factors such as average accommodation cost, flight availability and cost and currency exchange rate would distinguish destinations perceived as affordable from those seen as less affordable by travelers.

### 9.1.3 Research Objectives 3

To address this objective, we will use a two-way ANOVA. The independent variables are hotel brands and regions, and the dependent variable is guest satisfaction. We aim to determine whether guest satisfaction varies across different hotel brands and regions.

| Region        | Hotel Brands          |
|---------------|-----------------------|
| United States | JW Marriott           |
| China         | Sheraton              |
| Europe        | Courtyard by Marriott |
| Others        |                       |

Table 5: Classification of Variables

Assuming the data satisfies the necessary assumptions, SPSS will perform the two-way ANOVA.

#### Tests of Between-Subjects Effects

| Source                 | Type III<br>Sum of<br>Squares | df | Mean<br>Square | F | Sig. | Partial Eta<br>Squared |
|------------------------|-------------------------------|----|----------------|---|------|------------------------|
| Corrected<br>Model     |                               |    |                |   |      |                        |
| Intercept              |                               |    |                |   |      |                        |
| Region                 |                               |    |                |   | *    | #                      |
| Hotel Brands           |                               |    |                |   | **   | ##                     |
| Region*Hotel<br>Brands |                               |    |                |   | ***  | ###                    |
| Error                  |                               |    |                |   |      |                        |
| Total                  |                               |    |                |   |      |                        |
| Corrected<br>Total     |                               |    |                |   |      |                        |

Table 6: Two-way ANOVA Table from SPSS

We will focus on the red box to interpret the results. The hypotheses are:

|       | Main Effect                                                                                  |                                                                                            |                                                          |
|-------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------|
|       | Region                                                                                       | Hotel Brands                                                                               | Interaction Variable                                     |
| $H_0$ | There is no difference in the population <u>mean</u> of guest satisfaction among the regions | There is no difference in the population mean of guest satisfaction among the hotel brands | There is no interaction between region and hotel brands. |
| $H_1$ | Not all means are equal                                                                      | Not all means are equal                                                                    | There is interaction between region and hotel brands.    |

*Table 7: Summary of null and alternative hypotheses for 2-way ANOVA*

Reject  $H_0$  if p-value obtained in \*, \*\* and \*\*\* are less than 0.05. The Partial Eta Squared (X) value means that one specific factor can fully account for the changes in overall guest satisfaction, while the other several factors (represented by #, ##, and ### respectively) can't explain these changes.

#### 9.1.4 Research Objective 4

To address this objective, we will use one-way ANOVA. The independent variable is age group. The dependent variable is the sustainability awareness score. We would like to find out if sustainability awareness score differs between age groups which are 18-24 years old, 25-44 years old, 45 years old and above.

As the sample size is large enough, we can neglect the assumption of normality required in this statistical test and proceed directly to conduct Levene's test.

Variables to be used for test:

| Symbol     | Denotes                                                                              |
|------------|--------------------------------------------------------------------------------------|
| $\sigma_1$ | Population variance of sustainability awareness scores for 18-24 years old           |
| $\sigma_2$ | Population variance of sustainability awareness scores for 25-44 years old           |
| $\sigma_3$ | Population variance of sustainability awareness scores for 45 years old and above    |
| $\mu_1$    | Population <u>mean</u> of sustainability awareness scores for 18-24 years old        |
| $\mu_2$    | Population <u>mean</u> of sustainability awareness scores for 25-44 years old        |
| $\mu_3$    | Population <u>mean</u> of sustainability awareness scores for 45 years old and above |

Table 8: Variable denotation for one-way ANOVA

Output from SPSS:

##### Test of Homogeneity of Variances

|                                 |                                      | Levene Statistic | df1 | df2 | Sig. |
|---------------------------------|--------------------------------------|------------------|-----|-----|------|
| Sustainability awareness scores | Based on Mean                        |                  |     |     | *    |
|                                 | Based on Median                      |                  |     |     |      |
|                                 | Based on Median and with adjusted df |                  |     |     |      |
|                                 | Based on trimmed mean                |                  |     |     |      |

Table 9: Test of Homogeneity of Variances

### Levene's Test

$$H_0: \sigma_1 = \sigma_2 = \sigma_3$$

$$H_1: \text{Not all variances are equal}$$

Do not reject  $H_0$  if the value at \* > 0.05 (Table 8). This concludes that the population variances for sustainability awareness scores are not significantly different between age groups.

### **ANOVA**

|                | Sum of Squares | df | Mean Square | F | Sig. |
|----------------|----------------|----|-------------|---|------|
| Between Groups |                |    |             |   | *    |
| Within Groups  |                |    |             |   |      |
| Total          |                |    |             |   |      |

Table 9: ANOVA Table

### ANOVA Table

$$H_0: \mu_1 = \mu_2 = \mu_3$$

$$H_1: \text{Not all means are equal}$$

$H_0$  will be rejected if the value at \* < 0.05 (Table 9). This concludes that not all means are equal.

### 9.1.5 Research Objectives 6

Multiple Linear Regression (MLR) is used to address RO6. Let the predictor variables be exchange rate fluctuations, competitor hotel price and the inflation in traveler's home country. Meanwhile, the dependent variable will be the annual number of international bookings at Marriott hotels. The MLR function is:

$$\widehat{Amount} = \widehat{\beta}_0 + \widehat{\beta}_1 ERF + \widehat{\beta}_2 CHP + \widehat{\beta}_3 INFLATION + \widehat{\beta}_{41} TR_1 + \widehat{\beta}_{42} TR_2$$

$\widehat{Amount}$ : Estimated value of annual number of international bookings at Marriott hotels

$\widehat{\beta}_0$  : Intercept(Value of  $\widehat{Amount}$  when all the Xs are 0)

$\widehat{\beta}_k$ : When  $X_k$  increases by 1 unit,  $\widehat{Amount}$  increase by this value, with other independent variables remain constant; k=1,2,3,4

ERF: Exchange Rate Fluctuations

CHP: Competitor Hotel Price

INFLATION: Inflation In Traveler's Home Country

$TR_1, TR_2$ : Dummy Variable for Travel Restrictions

- 0=no restrictions(baseline)
- 1=partial restrictions
- 2=full restrictions

### Coefficients

| Model                                | Unstandardized Coefficients |            | Standardized Coefficients | t | Sig. | 95% Confidence Interval for $\beta$ |             |
|--------------------------------------|-----------------------------|------------|---------------------------|---|------|-------------------------------------|-------------|
|                                      | $\beta$                     | Std. Error | Beta                      |   |      | Lower Bound                         | Upper Bound |
| (Constant)                           |                             |            |                           |   |      |                                     |             |
| Exchange Rate Fluctuations           |                             |            |                           |   |      |                                     |             |
| Competitor Hotel Price               |                             |            |                           |   |      |                                     |             |
| Inflation In Traveler's Home Country |                             |            |                           |   |      |                                     |             |
| Travel Restrictions(partial)         |                             |            |                           |   |      |                                     |             |
| Travel Restrictions(full)            |                             |            |                           |   |      |                                     |             |

*Table 10: Coefficients table of Annual Number of International Bookings at Marriott Hotels from SPSS*

The estimated coefficient is derived from column B under unstandardized coefficient.

Partial T-test is used to test which variables have regression coefficients equal to 0 ( $\beta_i = 0$ ).

A variable is considered a significant predictor if the p-value is less than 0.05.

### Model Summary

| Model | R | R Square | Adjusted R Square | Std. Error of the Estimate |
|-------|---|----------|-------------------|----------------------------|
| 1     |   |          |                   |                            |

*Table 11: Model Summary from Model Fit from SPSS*

From “Model Summary” the  $R^2$  values measure the strength of association. The larger  $R^2$  shows that the model fits the observation better.

Global F-Test, along with the p-value, is used to test whether our model is significant in predicting the annual number of international bookings at Marriott hotels.



### ANOVA

|            | Sum of<br>Squares | df | Mean<br>Square | F | Sig. |
|------------|-------------------|----|----------------|---|------|
| Regression |                   |    |                |   | #    |
| Residual   |                   |    |                |   |      |
| Total      |                   |    |                |   |      |

Table 12: ANOVA for Global F-test from SPSS

### ANOVA

$$H_0: \beta_0 = \beta_1 = \beta_2 = \beta_3 = \beta_{41} = \beta_{42}$$

$$H_1: \text{Not all } \beta_s \text{ are 0}$$

If the p-value marked with # is less than 0.05, reject  $H_0$  and conclude that some of the independent variables can explain the variation in the dependent variables.

## 9.2 Qualitative

### 9.2.1 Research Objectives 5

From existing Marriott Bonvoy members, we will identify which program benefits are highly valued versus underutilized. Simultaneously, we aim to uncover perspectives from non-Bonvoy guests that may reveal overlooked barriers or motivations. Ultimately, this research will clarify evolving guest preferences to refine loyalty offerings that better align with customer priorities.

Each focus group will consist of one moderator and ten participants, limited to 90 minutes per session. Participants must be fluent in English and willing to openly discuss their experiences. All discussions will be recorded for internal analysis. The details have been previously discussed in Section 8.3.

The moderator will ask open-ended questions such as: “What motivated you to join the Bonvoy program?” and “Which rewards would make you stay loyal to Marriott versus competitors?”. Non-Bonvoy guests may be asked: “What would convince you to join a hotel loyalty program?”. Participants are also invited to raise relevant questions during the session.

## 10 Questionnaire

### 10.1 Main Questionnaire

### Questionnaire for Marriott's Market Research

Over the past few years, Marriott is being affected by pandemic. As life gradually returns to normal, we are making effort to recover. In order to more accurately meet your new needs and expectations in the new situation, we would like to consider your opinion by filling out this questionnaire which will take about 10 minutes. For your participation in the survey, you will receive a \$10 rebate voucher via e-mail upon completion of this questionnaire. Please answer them to the best of your abilities.

fionanyew@gmail.com [Switch accounts](#)

 Not shared 

\* Indicates required question

What is your age? (in years) \*

Your answer

What is your gender? \*

☐ Male

☐ Female

Which of the following is your permanent region of residence? \*

☐ United States

☐ China

☐ Europe

☐ Other:

Have you ever stayed at a Marriott hotel? \*

☐ Yes

☐ No

What is your average nightly accomodation cost? (in your local currency) \*

Your answer

What is your flight cost for trip? (in your local currency) \*

Your answer

How aware are you of Marriott hotels' sustainability initiatives? \*

0 1 2 3 4 5 6 7 8 9 10

Not at all ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ Extremely

Next

Clear form

Questionnaire for Marriott's Market Research

fionanyew@gmail.com Switch accounts Not shared \* Indicates required question

Marriott Customer

For respondents who answered "Yes" to having stayed at a Marriott hotel

Which of the following Marriott hotel brands have you stayed at? (Please select all that apply) \*

JW Marriott(for Luxury Brands)

Sheraton (for Premium Brands)

Courtyard by Marriott (for Select-Service Brands)

On average, how many international Marriott bookings do you make per year? (Numeric value) \*

Your answer

Which geographic region Marriott hotel you normally stayed at? \*

United States

China

Europe

Other:

How important is personalized service(e.g. room customization,technology integration) to you when choosing a hotel? \*

012345678910

Not at allExtremely

Rate flight availability for your destination. \*

12345678910

Extremely PoorExcellent

Rate the influence of currency exchange fluctuations on your travel decisions. \*

0 1 2 3 4 5 6 7 8 9 10

No Influence ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ Extremely Influential

Rate the overall affordability of Marriott hotels. \*

0 1 2 3 4 5 6 7 8 9 10

Not Affordable ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ Extremely Affordable

Rate your satisfaction with your most recent Marriott stay. \*

0 1 2 3 4 5 6 7 8 9 10

Extremely Dissatisfied ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ Extremely Satisfied

How would you classify the influence of exchange rate fluctuations on your booking decision? \*

- ☐ Not at all influential
- ☐ Slightly influential
- ☐ Moderately influential
- ☐ Very influential
- ☐ Extremely influential

How concerned are you about inflation in your home country affecting your travel plans? \*

- ☐ Not at all influential
- ☐ Slightly influential
- ☐ Moderately influential
- ☐ Very influential
- ☐ Extremely influential

How would you classify the impact of competitors' hotel prices on your booking decisions? \*

- ☐ Not at all influential
- ☐ Slightly influential
- ☐ Moderately influential
- ☐ Very influential
- ☐ Extremely influential

Would you be interested participating in a 90-minutes online focus group discussion regarding customer experience at Marriott hotels? (\$25 Amazon voucher will be given as an additional incentives) \*

- ☐ Yes
- ☐ No

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## 10.3 Questionnaire for non-Marriott Customer

Questionnaire for Marriott's Market Research

flonanyew@gmail.com [Switch accounts](#)

Not shared

\* Indicates required question

Non-Marriott Customer Questionnaire

(For respondents who answered "No" to having stayed at a Marriott hotel)

Which hotel brand(s) do you typically choose for international travel? \*

Your answer

How important is personalized service(e.g. room customization,technology integration) to you when choosing a hotel? \*

0 1 2 3 4 5 6 7 8 9 10

Not at all ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ Extremely

Rate the importance of the following factors when choosing a hotel (0 = Not Important at All, 10 = Extremely Important). \*

|                          | 1                     | 2                     | 3                     | 4                     | 5                     | 6                     | 7                     | 8                     |
|--------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Price Competitiveness    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Brand Reputation         | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Personalized Service     | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Sustainability Practices | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

Rate the following economic factors affecting your travel decisions (0 = No Impact, 10 = Strong Impact). \*

|                                            | 1                     | 2                     | 3                     | 4                     | 5                     | 6                     | 7                     | 8                     | 9                     |
|--------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Overall affordability of hotels you choose | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Impact of exchange rate fluctuations       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Concern about inflation affecting travel   | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Influence of competitors' hotel prices     | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

Would you be interested participating in a 90-minutes online focus group discussion regarding customer experience at Marriott hotels? (\$25 Amazon voucher will be given as an additional incentives)

☐ Yes

☐ No

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## 10.4 Interest Form for Focus Group Discussion

### Questionnaire for Marriott's Market Research

fionanyew@gmail.com [Switch accounts](#) 

 Not shared

\* Indicates required question

#### Focus Group Discussion Interest Form

Thank you for showing your interest in participating on Focus Group Discussion for customer experience at Marriott hotels. By filling up this form, you agree to be involved in the online focus group which would take around 90-minutes. The details will be emailed to you within 4 working days after you complete this form. The \$25 Amazon voucher incentive will be given to you via email upon finishing the online focus group discussion.

Name of Respondent \*

Your answer

Email address \*

Your answer

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## 11 Proposed Timeline Schedule and Budget Allocation Details

### 11.1 Timeline

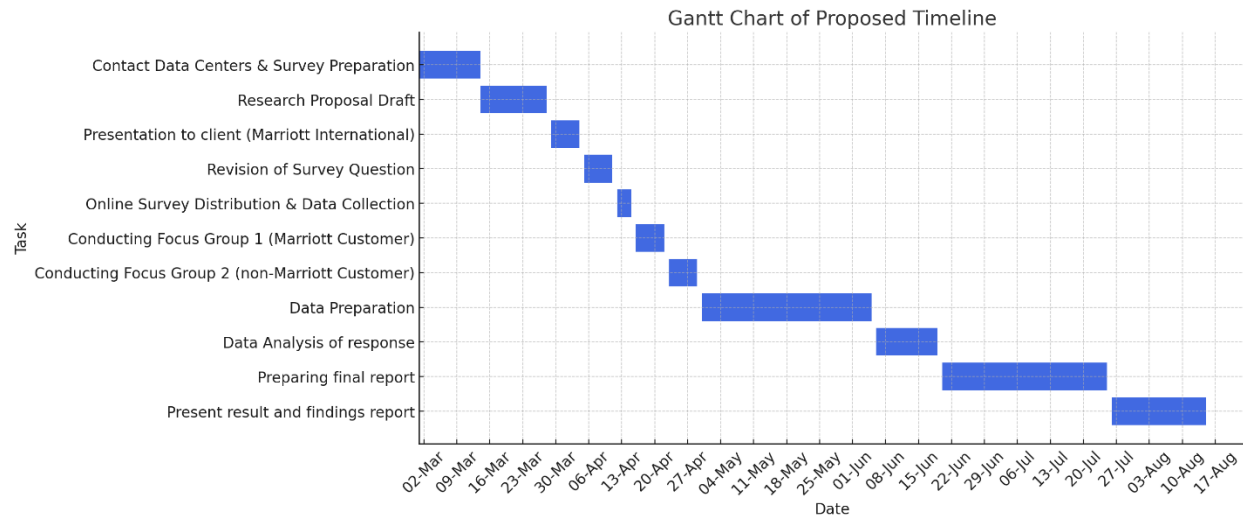


Figure 2: Gantt Chart of Proposed Timeline in 6 months

The project will start on March 1<sup>st</sup> and is expected to finish by August 1<sup>st</sup>. We will present the findings to Marriott International in August 14<sup>th</sup>.

### 11.2 Estimated Budget Allocation

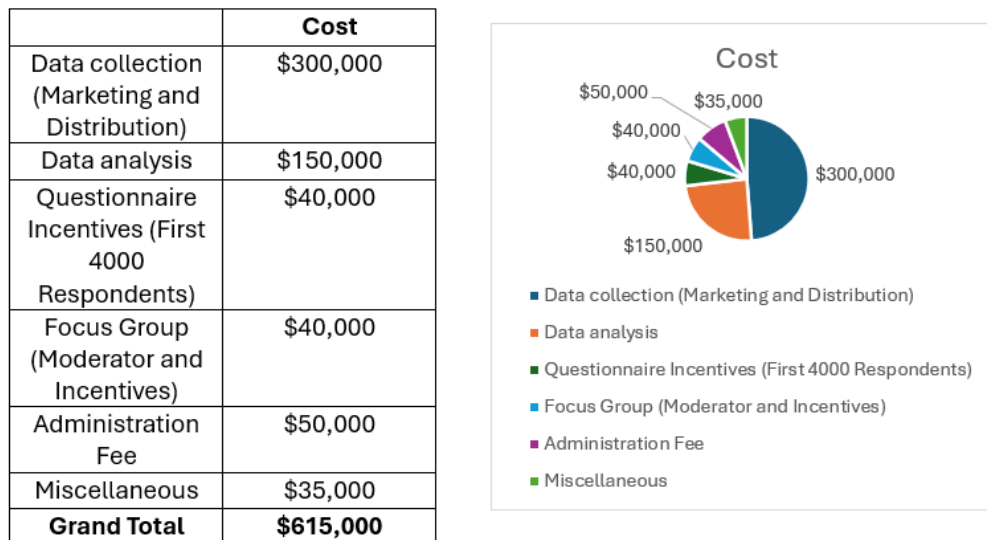


Figure 3: Costing table (left) and pie chart(right)

Data collection will be the largest portion of pie which approximately 49%, reflecting the importance of analyzing the collected data. \$615,000 will be the total cost of the project.



## 12 Further Recommendations

### 12.1 Government collaboration

Marriott must continuously monitor evolving health and safety regulations in key markets. For example, if governments mandate post-pandemic air quality certifications for public spaces, Marriott could pioneer enhanced HVAC (Heating, Ventilation, and Air Conditioning) system upgrades. Proactive compliance would strengthen brand trust and differentiate its properties as "certified-safe" destinations.

### 12.2 Personalized loyalty engagement

Tailored communication strategies could optimize Bonvoy membership appeal. Future campaigns should directly address hesitations uncovered in focus groups (e.g., "point expiration anxiety") while highlighting underutilized perks like exclusive event access. According to McKinsey & Company's 2023 analysis, effective personalization strategies can drive 10-15% revenue growth and 20-30% customer satisfaction improvements in service industries (McKinsey & Company, 2023).

## 13 Appendix

*2023 Annual Report*, marriott.gcs-web.com/static-files/b82978a6-9d28-4e38-9855-fc4ae2cebe11. [Accessed 25 Feb. 2025].

Abdey, J., 2021. ST3188 Statistical Methods for Market Research. [online] University of London. Available at: <<https://emfss.elearning.london.ac.uk/>> [Accessed 24 February 2025].

Abdey, J. and Chan, D., 2021. 3188 Statistical Methods for Market Research. [online] Singapore: Dion Chan. Available at: <<https://simconnect.simge.edu.sg/>> [Accessed 24 February 2025].

Meng-Jia Wu a, et al. “Response Rates of Online Surveys in Published Research: A Meta-Analysis.” *Computers in Human Behavior Reports*, Elsevier, 26 May 2022, [www.sciencedirect.com/science/article/pii/S2451958822000409](http://www.sciencedirect.com/science/article/pii/S2451958822000409). [Accessed 8 February 2025].

“Focus Groups.” *UX Research Methods for Discovery*, User Interviews, 15 Nov. 2024, [www.userinterviews.com/ux-research-field-guide-chapter/focus-groups](http://www.userinterviews.com/ux-research-field-guide-chapter/focus-groups). [Accessed 14 February 2025].

“What Is Personalization?” *McKinsey & Company*, McKinsey & Company, 30 May 2023, [www.mckinsey.com/featured-insights/mckinsey-explainers/what-is-personalization](http://www.mckinsey.com/featured-insights/mckinsey-explainers/what-is-personalization). [Accessed 19 February 2025].

### Image Used

Marriott International, Inc. “Marriott International Reports Third Quarter 2024 Results.” *PR Newswire: Press Release Distribution, Targeting, Monitoring and Marketing*, Cision PR Newswire, 4 Nov. 2024, [www.prnewswire.com/news-releases/marriott-international-reports-third-quarter-2024-results-302294817.html](http://www.prnewswire.com/news-releases/marriott-international-reports-third-quarter-2024-results-302294817.html).

| Requirement                                                                                                                                                                                 | Section     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Provide a full summary of the research brief, including the aims of the research.                                                                                                           | 1           |
| Demonstrate an understanding of the market or business context as well as any other publicly available research done in this area.                                                          | 1,2,3,4,5,6 |
| Detail how the fieldwork would be conducted, i.e. face-to-face, telephone, online, focus groups, mixed mode etc.                                                                            | 7,8         |
| Explain the proposed sampling method as well as other sampling methods considered, including details on any sampling frame to be used.                                                      | 8           |
| Detail the information that would be gathered and collected by the research.                                                                                                                | 8           |
| Explain how you would use any customer or operational data supplied to you by the client.                                                                                                   | 8           |
| Describe what multivariate analysis techniques you propose and how these would help the client's research aims. (You are not required to <u>actually conduct any analysis.</u> )            | 9           |
| Detail the proposed sample size necessary to construct confidence intervals around the survey estimates.                                                                                    | 8           |
| An appropriate questionnaire which would capture suitable data to perform the proposed multivariate analysis. (You are not required to <u>actually run the questionnaire in practice.</u> ) | 10          |
| Proposed further research, i.e. include ideas for how some business or <u>organizational</u> objectives might be helped by further and different research.                                  | 12          |

| Research Aims                                                                                                                                                                       | Research Questions                                                                                              | Research Objectives                                                                                                         | Hypothesis                                                                                                                                        | Statistical Techniques & Variables                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RA1- Understand customer preferences for personalized services, and technology integration, in order to deliver exceptional guest experience based on different demographic segment | Do customer preferences for personalized services differ between Marriott customers and non-Marriott customers? | To examine how customer preferences for personalized services differ between Marriott customers and non-Marriott customers. | There is a difference in preferences for personalised service between Marriott customers and non-Marriott customers.                              | Independent sample t-test<br>independent variable: Ycustomer type(dichotomous)<br>dependent variable: Ypreferences for                                                                                                                                                                                                                                                                               |
| RA2- Identify emerging trends and destination hotspots.                                                                                                                             | What factors distinguish between destinations perceived as affordable and those perceived as less affordable?   | Determine the key factors that influence travelers' perceptions of a destination's affordability.                           | The effect of factor significantly differentiate between destinations perceived as affordable and those perceived as less affordable.             | 3 group Discriminant Analysis<br>Categorical<br>Y-affordability(categorical)<br>*high affordable<br>*moderate affordable<br>*low affordable                                                                                                                                                                                                                                                          |
| RA3- Assess guest satisfaction across different brands and regions.                                                                                                                 | How do different hotel brands and regions affect guest satisfaction levels?                                     | To determine if guest satisfaction varies significantly across different hotel brands and regions                           | The effect of hotel brand on guest satisfaction varies across regions.                                                                            | 2 way ANOVAY-continuous<br>Guest satisfaction<br>X-categorical<br>X1-hotel brand (JW Marriott/Sheraton/Courtyard by Marriott)<br>X2-Different geo region (United States,china ,Europe)                                                                                                                                                                                                               |
| RA4- Explore the impact of sustainability initiatives on customer loyalty.                                                                                                          | Do different age groups have different levels of awareness about Marriott's sustainability initiatives?         | To compare sustainability awareness among different age groups.                                                             | There is a significant difference in sustainability awareness among different age groups.                                                         | One-way ANOVAY (continuous): sustainability awareness score<br>X (categorical): Age group (young, middle-aged, older)                                                                                                                                                                                                                                                                                |
| RA5- Evaluate the effectiveness of customer loyalty programs.                                                                                                                       | What are the key factors that influence customers to join the Bonvoy rewards program?                           | Identify the main reasons why people join the Bonvoy rewards program.                                                       |                                                                                                                                                   | focus group                                                                                                                                                                                                                                                                                                                                                                                          |
| RA7- Understanding how economic factors influence travel behaviour could help the company adapt to changing market conditions                                                       | How do economic factors influence the annual number of international bookings at Marriott hotels?               | To examine the impact of exchange rate fluctuations on the annual number of international bookings at Marriott hotels.      | Annual number of bookings is influenced by the following variables:<br>-exchange rate<br>-inflation<br>-competitor prices<br>-travel restrictions | MLR:Y (annual number of international bookings at Marriott hotels) → Continuous (total bookings)<br>X1: Exchange rate fluctuations (Continuous) - value of local currency vs. USD<br>X2: Inflation in travelers' home country (continuous)<br>X3: Competitor hotel prices (continuous)<br>X4: Travel restrictions (categorical) [ 0 = no restriction, 1 = partial restriction, 2 = full restriction] |